13	B	Work is done on the air through compression by the pump, increasing its internal energy. Since the bicycle pump is frictionless and well-insulated, there is no heat exchange with the surroundings. According to the first law of thermodynamics: $$\Delta U = Q + W$$ where • ΔU is the change in internal energy of the air • $Q = 0$ (no heat is transferred, as the pump is insulated) • $W > 0$ (work is done **on** the gas) Hence, $\Delta U = W$, the internal energy of the air increases, which results in a rise in thermodynamic temperature, as the thermodynamic temperature is directly proportional to internal energy for a gas.
14	D	Displacement-time graph of SHM